MLR Estimates Summary

We assume the k-regressors MLR model $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_{p-1} x_{p-1} + \epsilon$ with $\epsilon \overset{\text{iid}}{\sim} \mathrm{N}(0, \sigma^2)$, with data denoted by

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1,p-1} \\ 1 & x_{21} & x_{22} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{n,p-1} \end{bmatrix}_{n \times p}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{p-1} \end{bmatrix}_{p \times 1}, \text{ and } \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

- The OLS estimator is $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \sim \mathbf{N}\big(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\big)$. For a specific one, $\hat{\beta}_1 \sim \mathrm{N}(\beta_1, [\sigma^2(\mathbf{X}'\mathbf{X})^{-1}]_{11})$, where $[\sigma^2(\mathbf{X}'\mathbf{X})^{-1}]_{11}$ denotes the $(1,1)$-entry.
- The fitted values are $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. We can write $SS_{Res} \coloneqq \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}$.
- The noise variance is estimated by $\hat{\sigma}^2 = SS_{Res}/(n-p)$ which is also named MS_{Res}.

Interested Quantity	**Estimate**	**Std. Error**	**Test** $H_0: \beta_j = a$	**Interval Estimate**	**Code**
Any specific β_j	j-th element of $\hat{\boldsymbol{\beta}}$: $\hat{\beta}_j = \left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}\right]_j$	(j,j)-entry of $\sqrt{\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}}$: $\mathrm{se}(\hat{\beta}_j) = \sqrt{[\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$	$\dfrac{\hat{\beta}_j - a}{\mathrm{se}(\hat{\beta}_j)} = t \overset{H_0}{\sim} t_{n-p}$	$\hat{\beta}_j \pm t_{n-p}(\alpha/2)\,\mathrm{se}(\hat{\beta}_j)$	`confint(reg, parm="x[j]")`

- At a new $\mathbf{x}_0$, the CI for the mean response is $\mathbf{x}_0'\hat{\boldsymbol{\beta}} \pm t_{n-p}(\alpha/2)\sqrt{\hat{\sigma}^2\mathbf{x}_0'(X'X)^{-1}\mathbf{x}_0}$ / the PI for a new observation is $\mathbf{x}_0'\hat{\boldsymbol{\beta}} \pm t_{n-p}(\alpha/2)\sqrt{\hat{\sigma}^2[1 + \mathbf{x}_0'(X'X)^{-1}\mathbf{x}_0]}$
- The following `R output` serves as an illustration.

```
> summary(reg)
Call:
lm(formula = y ~ x1 + x2, data)
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.478742 10.244808   0.144   0.8865
x1           -3.201330  0.950000  -3.370   0.0023 **
x2            5.375050  1.200000   4.479   0.0001 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 3.028 on 27 degrees of freedom
Multiple R-squared:  0.8,    Adjusted R-squared:  0.7704
F-statistic:   41.24 on 4 and 27 DF,  p-value: 4.32e-09
```

The fitted regression line is $\hat{y} = 1.4787 - 3.2013x_1 + 5.3751x_2$. For the specific β_2:

- the OLS estimate is $\hat{\beta}_2 = [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}]_2 = 5.375$
- the noise has estimated $\hat{\sigma} = \sqrt{SS_{Res}/(n-p)} = 3.028$.
- the estimated standard error is $\mathrm{se}(\hat{\beta}_2) = \sqrt{[\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}]_{22}} = 1.20$
- the test statistic for $H_0: \beta_2 = 0$ is

$$\frac{\hat{\beta}_2 - 0}{\mathrm{se}(\hat{\beta}_2)} = \frac{5.375}{1.20} = 4.479 \overset{H_0}{\sim} t_{n-p} \text{ with } p\text{-value} = 0.0001.$$

- Other β can be explained in a similar way.

The F test will be introduced in the next page `MLR ANOVA Table`.

MLR ANOVA Table

For the MLR $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, we can group the first $p-r$ regressors $\boldsymbol{\beta}_1 := (\beta_0, \cdots, \beta_{p-r-1})'$ and the last r regressors $\boldsymbol{\beta}_2 := (\beta_{p-r}, \cdots, \beta_{p-1})'$. Specifically,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} = \left[\begin{array}{cccc:cccc} 1 & x_{11} & \cdots & x_{1,p-r-1} & x_{1,p-r} & \cdots & x_{1,p-1} \\ 1 & x_{21} & \cdots & x_{2,p-r-1} & x_{2,p-r} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{n,p-r-1} & x_{n,p-r} & \cdots & x_{n,p-1} \end{array}\right] \left[\begin{array}{c} \beta_0 \\ \vdots \\ \beta_{p-r-1} \\ \hdashline \beta_{p-r} \\ \vdots \\ \beta_{p-1} \end{array}\right] + \boldsymbol{\epsilon} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\epsilon}$$

where the first four columns form $\mathbf{X}_1$ and the last three columns form $\mathbf{X}_2$.

To test $H_0 : \boldsymbol{\beta}_2 = 0$, i.e. all last r β's are 0 or H_1 : some of them is not 0, we do the following procedure:

- First build the full model $\hat{\mathbf{y}}_{\text{full}} = \mathbf{X}\hat{\boldsymbol{\beta}}$, and compute the full model $SS_R(\boldsymbol{\beta})$ and $MS_{Res}(\boldsymbol{\beta})$
- Then under H_0, build the reduced model $\hat{\mathbf{y}}_{\text{red}} = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1$, and compute the reduced model $SS_R(\boldsymbol{\beta}_1)$.
- The difference $SS_R(\boldsymbol{\beta}) - SS_R(\boldsymbol{\beta}_1) =: SS_R(\boldsymbol{\beta}_2|\boldsymbol{\beta}_1)$ measures the extra-sum-of-squares in regression of $\boldsymbol{\beta}_2$.
- Under $H_0 : \boldsymbol{\beta}_2 = 0$, we can derive the partial F-test statistic:

$$F = \frac{SS_R(\boldsymbol{\beta}_2|\boldsymbol{\beta}_1)/r}{MS_{Res}(\boldsymbol{\beta})} \overset{H_0}{\sim} F_{r,n-p}.$$

The following `R output` is an illustration of the sequential ANOVA.

```
> anova(reg)
Analysis of Variance Table

Response: y
          Df  Sum Sq Mean Sq F value Pr(>F)
x1         1  220.50  220.50   18.40 0.0002 ***
x2         1   96.80   96.80    8.08 0.0083 **
Residuals 27  323.46   11.98
```

The full model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ with $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)'$ has $SS_{Res}(\boldsymbol{\beta}) = 323.46$ and $MS_{Res}(\boldsymbol{\beta}) = 11.98$.

(1) The `x1`-row compares $\boldsymbol{\beta}_1 = (\beta_0)'$ vs $\boldsymbol{\beta}_2 = (\beta_1)'$ to test $H_0 : \beta_1 = 0$:

$$SS_R(\boldsymbol{\beta}_2|\boldsymbol{\beta}_1) = 220.50 \;\Rightarrow\; F = \frac{SS_R(\boldsymbol{\beta}_2|\boldsymbol{\beta}_1)/r}{MS_{Res}(\boldsymbol{\beta})} = \frac{220.50/1}{11.98} = 18.40$$

(2) The `x2`-row compares $\boldsymbol{\beta}_1 = (\beta_0, \beta_1)'$ vs $\boldsymbol{\beta}_2 = (\beta_2)'$ to test $H_0 : \beta_2 = 0$:

$$SS_R(\boldsymbol{\beta}_2|\boldsymbol{\beta}_1) = 96.80 \;\Rightarrow\; F = \frac{SS_R(\boldsymbol{\beta}_2|\boldsymbol{\beta}_1)/r}{MS_{Res}(\boldsymbol{\beta})} = \frac{96.80/1}{11.98} = 8.08.$$

To test overall $H_0 : \beta_1 = \beta_2 = 0$ (the whole regression is insignificant), we set $\boldsymbol{\beta}_1 = (\beta_0)'$ and $\boldsymbol{\beta}_2 = (\beta_1, \beta_2)'$. This is the `F-statistic` reported by `summary(reg)`, which can also be computed from the above `anove(reg)` table. (Try yourself.)